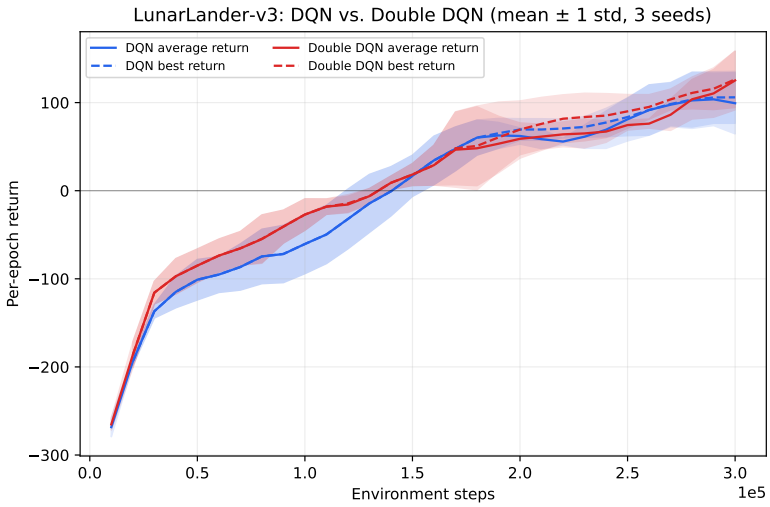
Assignment 3: Q-Learning and Actor-Critic Algorithms

Andrew ID: `jingshup`
Collaborators: None.
NOTE: Please do **NOT** change the sizes of the answer blocks or plots.

1 Question 1: DQN and DDQN (LunarLander-v3) - [7.5 points]

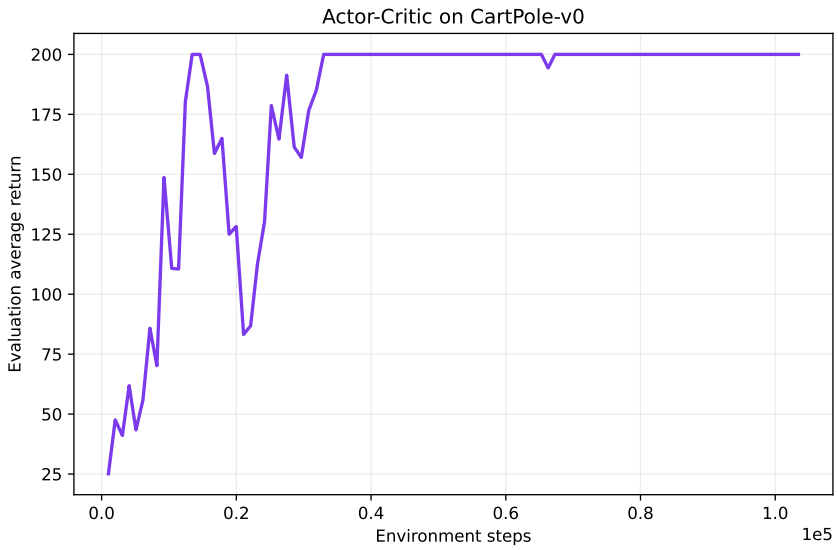
Question 1



Mean performance across seeds 1–3; shading shows one standard deviation. Both methods use $\gamma = 0.99$, target updates every 2,000 steps, and epsilon decay from 1.0 to 0.02 over the first 90,000 steps. At 300,000 steps, DDQN averages 125.2 return versus 99.3 for DQN across three seeds.

2 Question 2: Actor-Critic (CartPole-v0) - [3.75 points]

Question 2



Actor-critic reaches the CartPole-v0 maximum return of 200.

3 Question 3: Actor-Critic (InvertedPendulum-v4) - [3.75 points]

Question 3



Evaluation return for actor-critic with 10 target updates and 10 gradient steps per target.